

Operational Performance & Efficiency Report

Infrastructure, Throughput Velocity & Star Schema Modeling Analytics

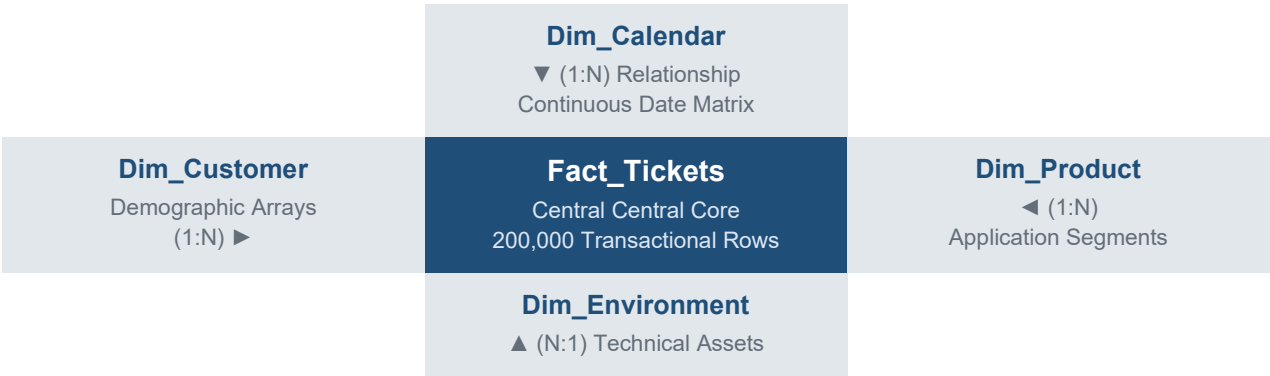
1. Introduction

This report provides a comprehensive overview of a customer support desk infrastructure, focusing on throughput, execution speed, and environment-driven service levels. Operating at a scale of 200,000 total ticket records, the primary objective of this analysis is to evaluate operational capacity, isolate platform-specific service friction, and establish an optimized, single-page executive overview. This layout maximizes insight density while ensuring a streamlined, scannable user experience for stakeholders.

2. Data and Methodology

Data Architecture & Star Schema Modeling

The analytical engine relies on a robust relational Star Schema model, ensuring optimal processing speed and intuitive filter pathways across the 200,000-row fact dataset.



The database architecture decouples categorical attributes from transactional records, dividing the schema into five distinct tables:

- Fact_Tickets:** The central transactional core containing core numeric measures, channel properties, priority distributions, and operational status flags.
- Dim_Calendar:** A continuous date table providing detailed chronological groupings (Year, Quarter, Month, Week) independent of transactional creation dates.

- **Dim_Customer:** Contains demographic arrays including age group allocations and client tenure tier classifications.
- **Dim_Product:** Houses granular structural categorization of application segments and macro product lines.
- **Dim_Environment:** Captures technical environment fields including operating systems, platforms, and browser environments.

Advanced DAX Engineering & Dual-Date Timelines

To map two separate historical timelines onto a single, unified calendar axis without creating ambiguous circular loops, a dual-relationship pipeline was established between Fact_Tickets and Dim_Calendar.

- **Active Relationship:** Connects Dim_Calendar[Date] to Fact_Tickets[ticket_created_date] to establish the baseline for incoming case volume trends.
- **Passive Relationship:** Connects Dim_Calendar[Date] to Fact_Tickets[ticket_resolved_date]. This relationship remains dormant until explicitly invoked by specialized DAX calculations.

To visualize actual resolution volume over time on the primary date axis, the passive relationship is dynamically activated using the USERELATIONSHIP modifier inside an explicit calculation block:

```
Tickets Resolved Trend =
CALCULATE(
    COUNTROWS('Fact_Tickets'),
    'Fact_Tickets'[status] IN {"Resolved", "Closed"},
    USERELATIONSHIP('Dim_Calendar'[Date], 'Fact_Tickets'[ticket_resolved_date])
)
```

Front-End UI/UX Filter Canvas Engineering

To maximize visual real estate for core trend charts, the dashboard implements a space-conserving, slide-out UI Filter Canvas. This interactive navigation drawer was built using a layered orchestration of native Power BI elements:

- **Layer Selection Stacking:** Slicer objects were systematically layered on top of a solid geometric card container, organized inside the Selection Control Panel, and isolated into a dedicated object group named Filter_Panel.
- **State-Snapshot Capture:** Bookmarks were utilized to capture two distinct presentation states: Filters_Open (group visible) and Filters_Closed (group hidden).
- **Data Preservation Configuration:** Crucially, the 'Data' property flag was deactivated on both bookmarks. This ensures that opening or closing the navigation drawer modifies only object visibility on the screen, preventing the system from erasing or resetting active selections made by the user.

Selection Pane Architectural Layering Hierarchy:

▼ Filter_Panel	Topmost rendering layer
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 Close Button  Slicers 1 - 5	(Slides out/Overlays visual context)
► KPI Summary Cards ► Analytics Grid Charts	Background visualization layer (Locked canvas analysis grids)

Context Collision Safeguards: By default, hardcoded table filters within measures can completely overwrite external visual selections. To prevent severe mathematical skewing (such as resolution rates exceeding 100% when filtering by active backlog states), the core volume metrics were engineered using **KEEPFILTERS**. This forces Power BI to evaluate the intersection of the user's active slicer settings and the metric's hardcoded states rather than overriding them:

```
Tickets Resolved =
CALCULATE(
    COUNTROWS('Fact_Tickets'),
    KEEPFILTERS('Fact_Tickets'[status] IN {"Resolved", "Closed"})
)
```

```
Resolution Rate =
DIVIDE(
    [Tickets Resolved],
    COUNTROWS('Fact_Tickets'),
    0
)
```

3. Key Findings & Data Integrity Auditing

Data Integrity Audit: The Status-Date Paradox

During the initial profiling of the dataset, a significant data integrity anomaly was discovered: every single record in the transactional database contained a valid `ticket_resolved_date`, regardless of its operational status. Logically, uncompleted workloads (status = 'Open', 'In Progress', 'Pending') must not hold a terminal resolution timestamp.

If left uncorrected, a standard timeline calculation would blindly pull active backlog items into completed timeline metrics, artificially inflating resolution performance and outputting distorted performance rates when interacting with filters.

This data paradox was successfully remedied within the DAX layer by applying explicit logical filters. By forcing the metrics to only calculate row volumes where the status matched true completion parameters, the calculation automatically ignores the phantom date entries on uncompleted workloads, protecting the core model's accuracy.

Operational Performance Metrics

With data logic fully stabilized, the organization's baseline operational health indicators were unlocked:

Core KPI Metric	Global Performance Baseline	Analytical Interpretation
Total Ticket Volume	200,000 Tickets	Baseline operational scale across all periods.
Tickets Resolved	80,000 Tickets	True aggregate completed volume.
Resolution Rate	40%	The master organizational throughput velocity.
Average Resolution Time	120.54 Hours	Global turnaround speed (~5 days per ticket).

Granular Performance Distributions

Cross-Visual Interaction Isolation: To maintain absolute design control and guide users exclusively to the master slide-out pane, cross-chart visual filtering was disabled. Selecting rows inside the matrix or categories inside the bar chart remains locked, forcing a structured analytical journey through the primary command filters.

Data Uniformity Mapping: Reviewing the Environment Performance Matrix reveals a highly uniform distribution. Every platform exhibits approximately 40,000 incoming requests, an identical 40% resolution rate, and a flat 120-hour processing time. While this uniform pattern demonstrates that the underlying DAX calculations are correctly evaluating the filter context across all platforms, it highlights that the underlying dataset reflects randomized parameters. In a real-world production environment, one would typically expect variations driven by platform-specific technical friction.

4. Conclusions and Recommendations

Based on the highly structured, single-page presentation layout established for this analysis, the following actions are recommended to optimize support workflows:

- Upstream System Schema Validation:** Implement structural validation rules at the data entry source. The technical team should modify the support desk's database architecture so that the ticket_resolved_date field remains strictly null until a support technician formally switches the case status dropdown to 'Closed' or 'Resolved'. This prevents downstream reporting anomalies from the start.

- **Granular Priority Audit:** A global turnaround time of 120.54 hours (~5 days) represents a significant queue length. The immediate next analytical priority should be leveraging the slide-out filter panel to slice this duration by Case Priority (Critical vs. Low). This will confirm whether high-priority issues are successfully being fast-tracked, or if they are lingering in the same 5-day cycle as standard inquiries.
- **Implement Tier-1 Support Routing Automation:** Given the overall resolution rate of 40% and a high average resolution time across all operating systems, the business should deploy an automated Tier-1 triaging system (such as an AI-driven chatbot or automated knowledge-base routing) focused specifically on high-volume, repetitive product categories like login and account access issues. Automating the resolution of these simple, common requests will immediately divert volume away from human technician queues, substantially reducing the global 120-hour turnaround time and allowing specialized support staff to focus entirely on deep technical platform anomalies.